



TA : Amany Sherif

Book Orbit

A .NET Clean Architecture BackEnd System.

Connecting students through the
orbit of knowledge sharing.



Team Members

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The Problem & Why It's Hard



Financial Waste

Students waste money on textbooks by abandoning them after a single semester



State Complexity

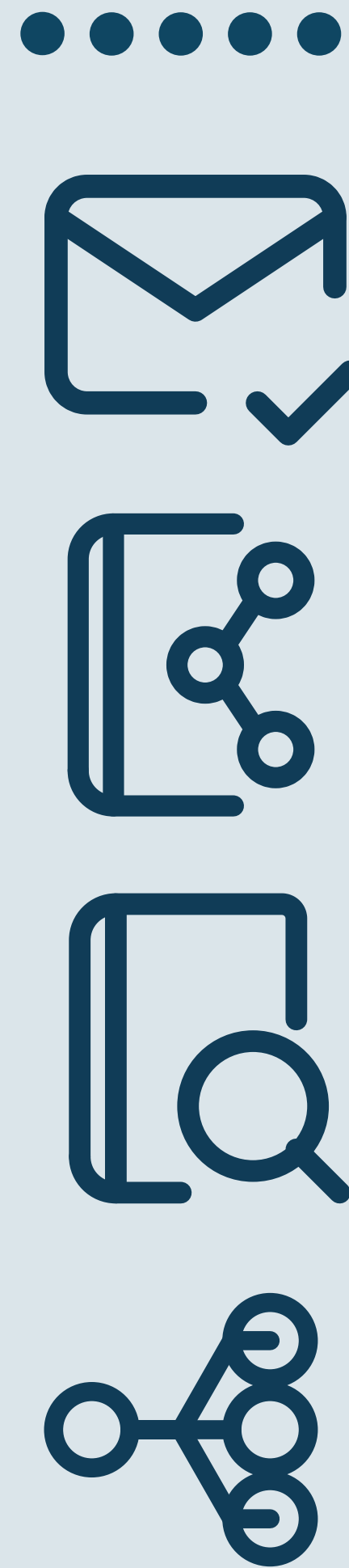
Peer-to-peer lending requires strict transaction state integrity — one missed transition breaks the entire chain



Zero Trust & Identity Mandate

No trusted, centralized platform exists for secure student-to-student lending. Identity trust is non-negotiable — only verified university students can participate in the ecosystem.

The Solution: BookOrbit at a Glance



- **Verify Email & Admin Approve**

Secures the platform through admin-gated onboarding utilizing mandatory university email verification.

- **Shared Catalog**

Separates the shared Book catalog from physical BookCopy ownership to enforce a single source of truth.

- **Browse & Request**

Allows students to list physical book copies while others browse, request, and manage borrowing lifecycles.

- **State Transitions**

Manages the complete borrowing lifecycle through strict states: pending, accepted, rejected, or cancelled.



System Architecture — Clean Architecture

- **Api**

(Controllers,

Middleware)

- **Infrastructure**

(EF Core, Identity,

JWT)

- **Application**

(CQRS, MediatR,

DTOs)

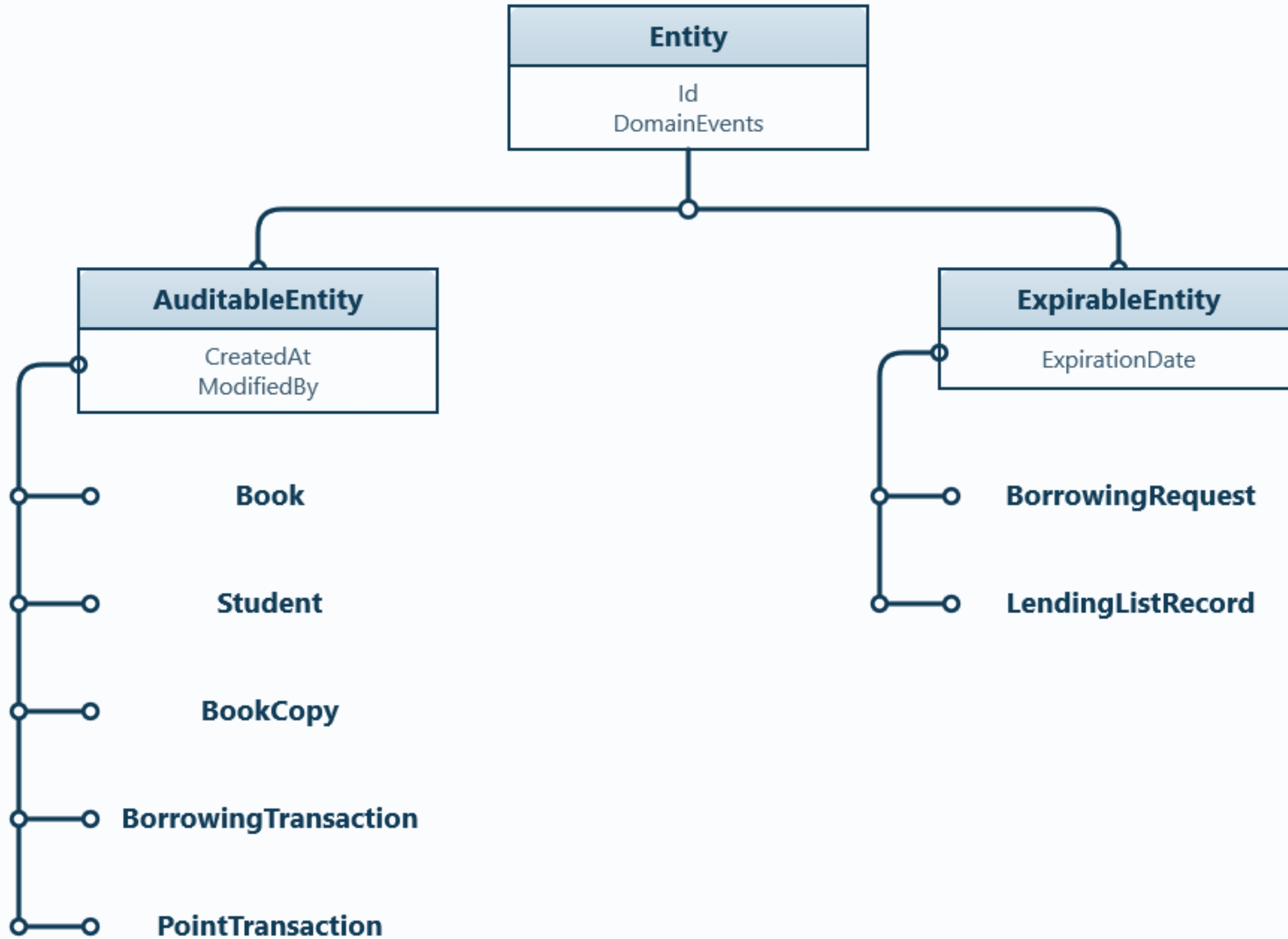
- **Domain**

(Entities, Value Objects,

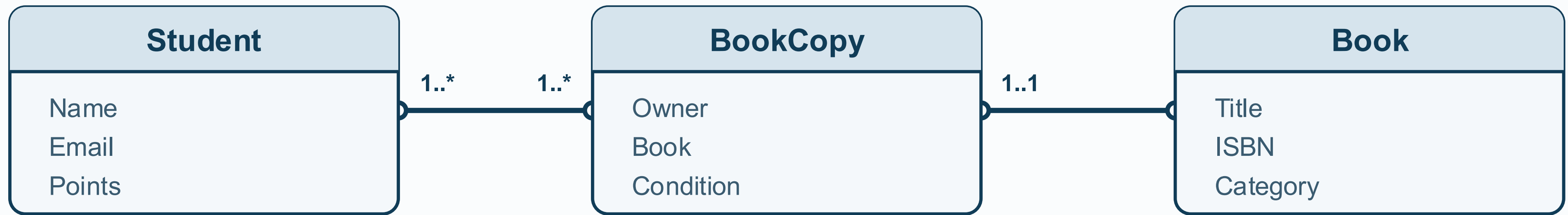
Events)



Domain Layer — Entity Hierarchy



Domain Entities — Student, Book & BookCopy



- **Student:** Points economy, verified identity.
- **Book:** Master catalog entity.
- **BookCopy:** Physical copy tracking condition.

The Lending & Borrowing Workflow

- **List**

(LendingListRecord)
LendingList is a physical book.

- **Shared**

Catalog
Separates the shared Book catalog from physical BookCopy ownership to enforce a single source of truth.

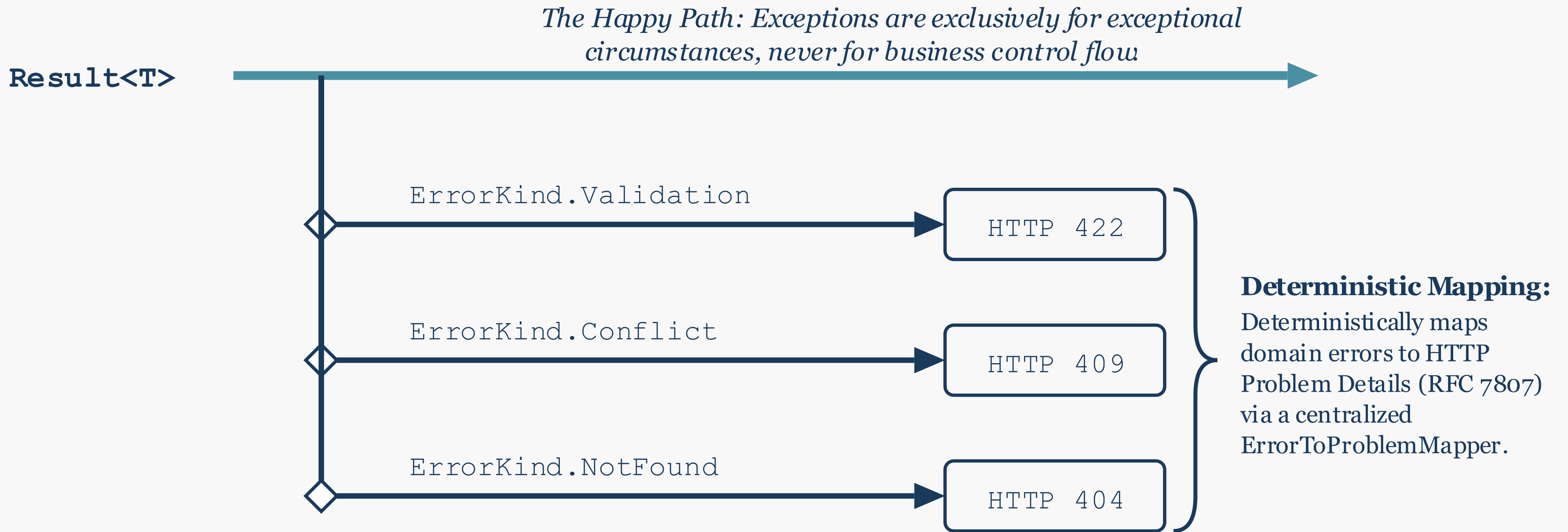
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- **State**

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Exception-less Control Flow: Railway-Oriented Programming



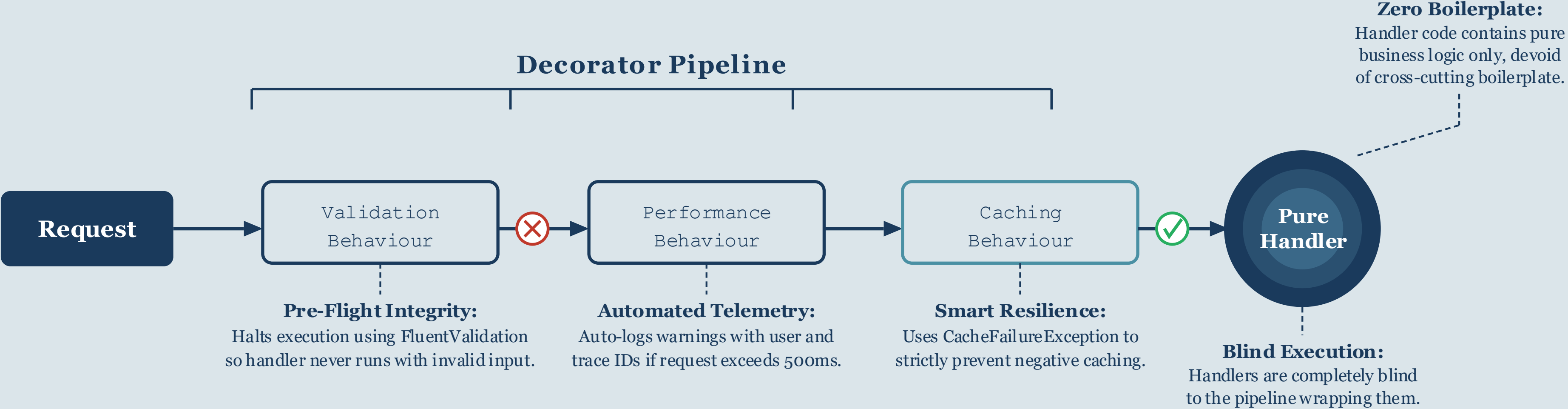
- **Forced Acknowledgment**

Forces the Presentation layer to explicitly acknowledge and handle every potential failure path.

- **No Hidden Modes**

Delivers a 100% predictable API contract that eliminates unpredictable bubbling and hidden failure modes.

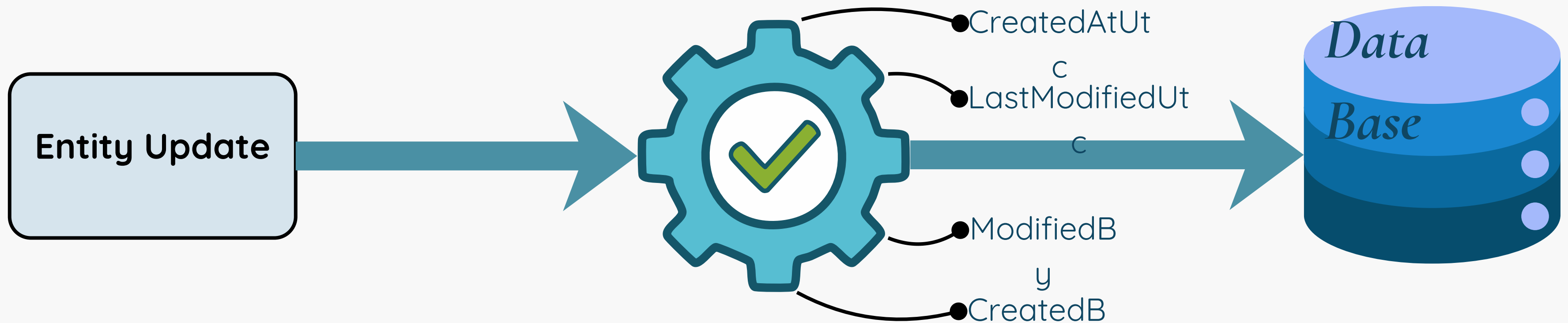
Application Pipeline: MediatR Behaviors



Pre-Flight Integrity:
Halts execution using
FluentValidation so the handler
never runs with invalid input.

Smart Persistence: EF Core Interceptors

Global Hook: Hooks directly into `SaveChangesAsync` via a globally registered `AuditableEntityInterceptor`.

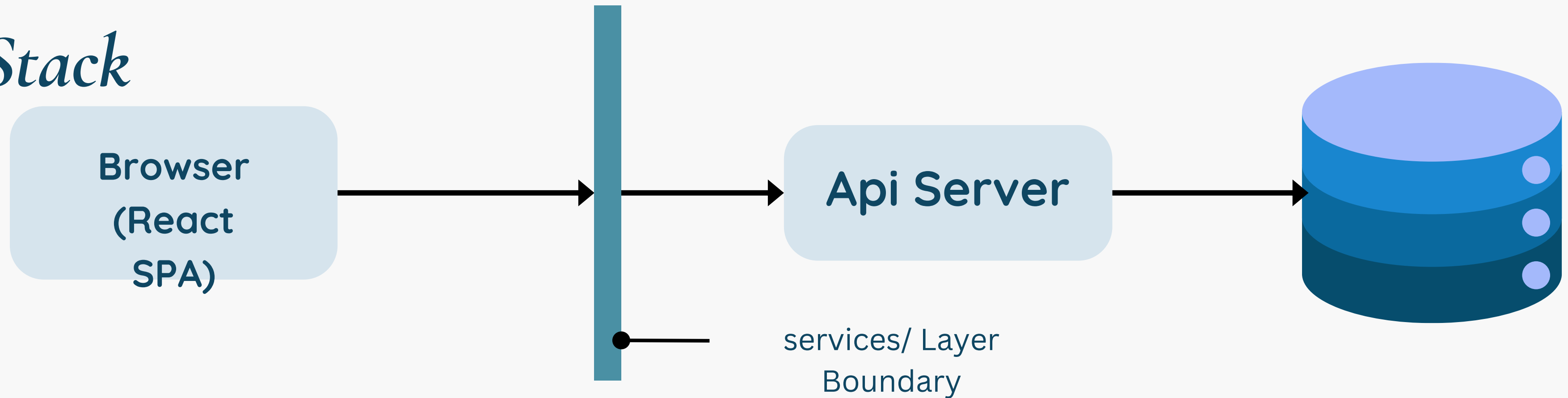


Automatic Injection: Automatically injects timestamps and user data on every single entity state change.

Transactional Consistency: Dispatches Domain Events via MediatR strictly before the database commit to guarantee transactional consistency.

Frontend Architecture & Tech

Stack



React v19 & Tailwind CSS v3.4 deliver a highly responsive, utility-driven Single Page Application.

React Router v7 handles all client-side navigation and protected route enforcement.

Framer Motion & Lucide React provide micro-interactions and consistent iconography without bloat.

Feature-based architecture organizes the src/ directory by domains like pages/auth/ and pages/lending/.

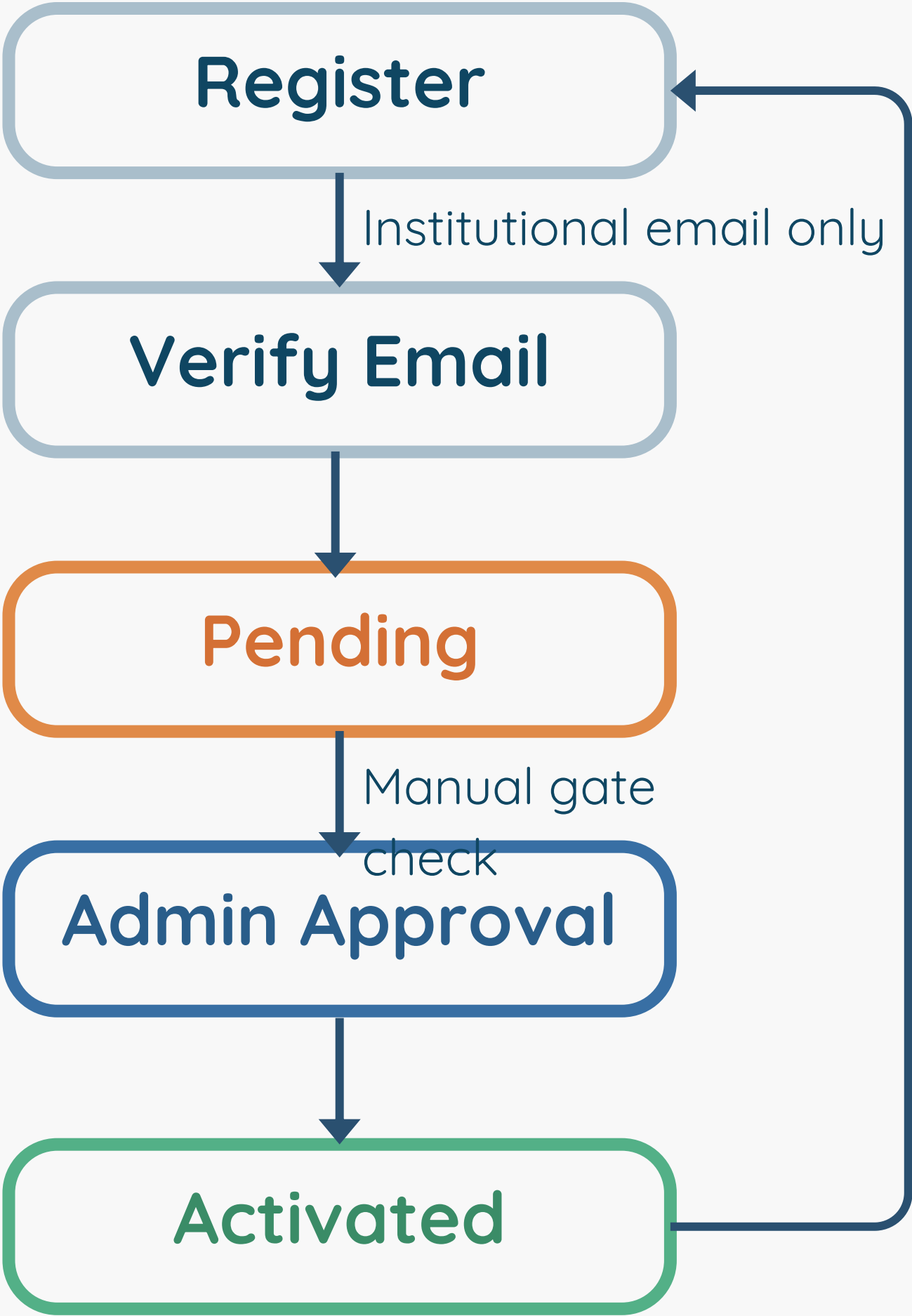
Service layer isolation ensures the UI never directly calls fetch, routing all API traffic through services/.

Testing Library enforces reliability across critical UI components and custom hooks.

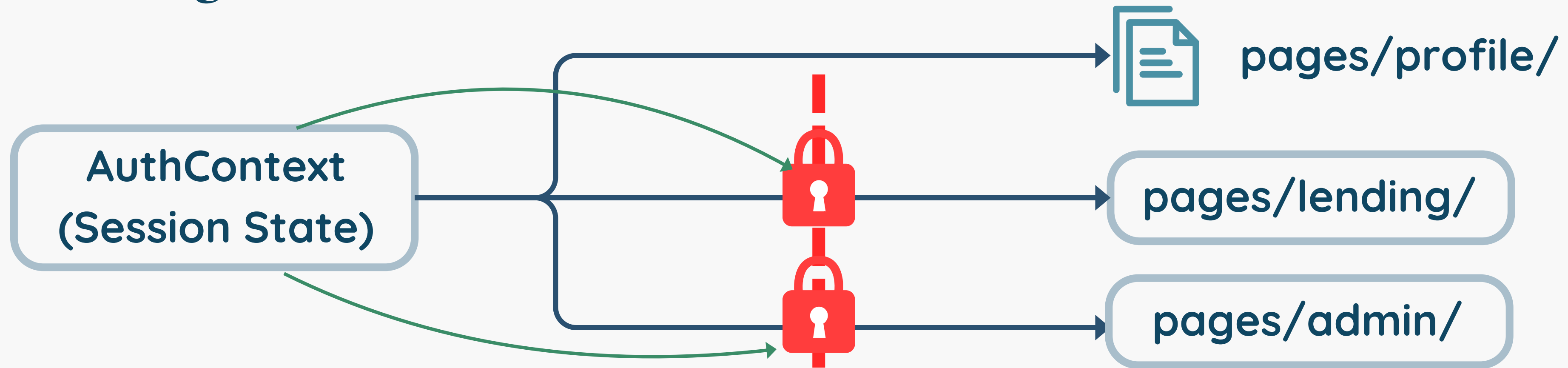
Authentication Flow & Session

Management

- **University email gating**
strictly limits registration to verified institutional domain addresses.
- **JWT dual-token strategy**
rotates the short-lived access token from the secure, long-lived refresh token.
- **Silent refresh**
runs automatically via AuthContext, ensuring users are never interrupted mid-session.
- **Environment keys**
securely isolate token identifiers away from the compiled application bundle.
- **Manual gatekeeping**
requires Admin approval to finally transition an account to activated.



Routing & State Architecture



Client-side routing via React Router v7 prevents full-page reloads while managing complex navigational state.

Guard checkpoints ensure no protected view renders without first passing a strict authentication check.

Single source of truth is securely maintained by AuthContext for all session data across the application.

API boundary enforcement routes everything through services/, creating a single point of configuration.

Custom hooks extract complex business logic out of UI components into reusable, isolated functions.

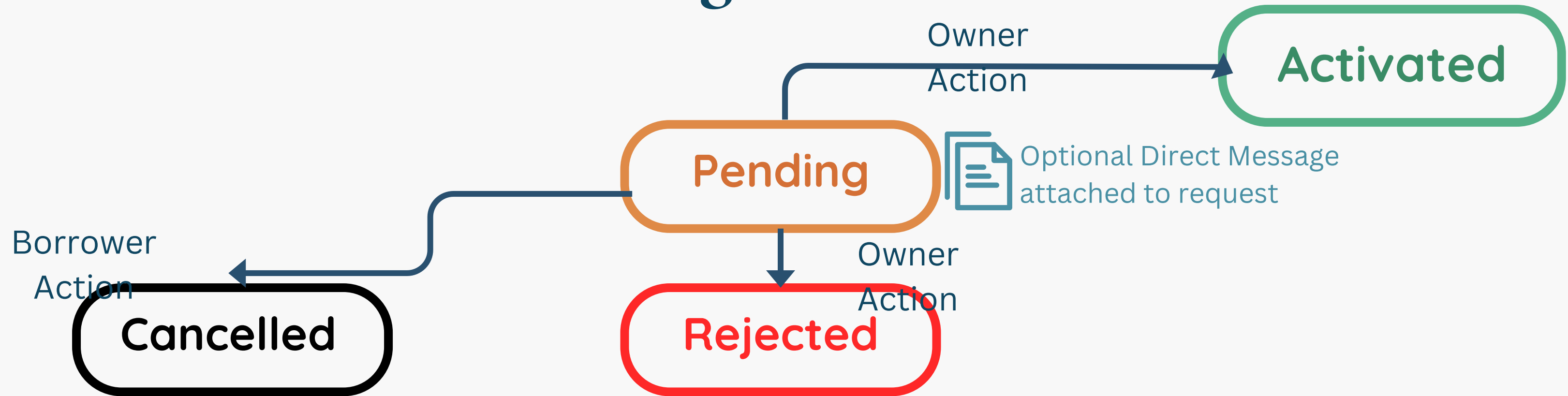
Centralized configuration makes adapting to backend URL changes a single-line adjustment.

Core User Flow: Listing a

- **Book-first architecture**
requires students to search the shared database before creating new entries.
- **Physical copy attachment**
securely links a user's specific physical book to the centralized master record.
- **New entry suggestions**
trigger when a book is missing, pushing the data to an admin review queue.
- **Asynchronous approval**
prevents unverified or duplicate books from cluttering the shared market.
- **Explicit listing action**
formally moves a verified physical copy into the active lending market.



Core User Flow: Borrowing a Book



Faceted browsing allows students to filter the lending market by title, category, and real-time availability.

Asynchronous requests allow borrowers to send targeted pings with optional direct messages.

Instant notification alerts the physical owner that their actively listed copy has a prospective borrower.

State transition controls grant the owner exclusive rights to move a request from pending to accepted or rejected.

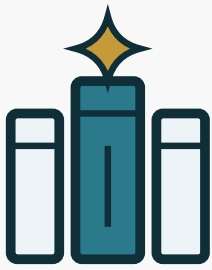
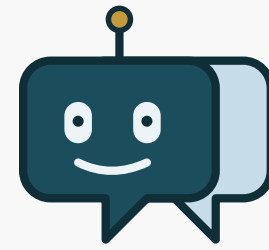
Proactive cancellation allows the borrower to abort the request unilaterally before an owner responds.

Physical handover occurs off-platform only after the accepted state is mutually confirmed.

Admin Panel & Platform Control

Permission	Standard Student Role	Administrator Role
• Browse Lending Market	✓	✓
• Access pages/admin/routes	✗	✓
• Approve New Book Suggestions	✗	✓
• Ban / Unban User Accounts	✗	✓
• Isolated administrative domain securely houses all governance tools exclusively under the pages/admin/directory endpoints.		
• Route-level authorization strictly blocks standard students from accessing any administrative endpoints.		
• Granular account moderation empowers admins to approve, reject, ban, or unban user profiles.		
• Route-level authorization strictly blocks standard students from accessing any administrative endpoints.		
• Catalog hygiene enforcement routes all new book suggestions through a manual admin approval gate.		

Future Program Roadmap

- *RECOMMENDATION* 
- *AI OPTIMIZES IMAGES OF BOOKS CAPTURED.* 
- *AI-POWERED SEARCH: USER DOESN'T HAVE TO SEARCH BY EXACT NAME.* 
- *INTEGRATE AN AI CHATBOT.* 



Thank you!

